Governing Equations for MEA Post-Combustion CO_2 Capture Simulation

Course Report
ChE206 — Separation Processes

Subject:	Equations, Models, and Derivations of a Rate-Based MEA Absorber–Stripper Cycle
Solvent:	30 wt% Monoethanolamine (MEA) in Water
Process:	Post-Combustion CO_2 Capture from Flue Gas
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1 Introduction and Process Overview

Post-combustion CO_2 capture using monoethanolamine (MEA) is the most commercially mature solvent-based carbon capture technology. The process consists of two thermally coupled packed columns operating at near-atmospheric pressure: an **absorber**, in which flue gas rising counter-currently contacts lean MEA solution and CO_2 is chemically absorbed; and a **stripper** (regenerator), in which the CO_2-rich solvent is heated in a reboiler to reverse the absorption reactions and regenerate lean solvent for recycle.

The simulation implemented here is a *rate-based* (non-equilibrium) model, meaning mass transfer rates — not equilibrium stage efficiencies — govern the column performance. This report derives all governing equations from first principles: two-film mass transfer theory, reaction kinetics, chemical equilibrium (VLE), energy balances, and the operating line material balance.

1.1 Column Indexing Convention

Location	Gas	Liquid
Stage $j = 0$	Inlet (bottom)	Outlet (bottom)
Stage $j = N - 1$	Outlet (top)	Inlet (top)

Gas flows bottom $\rightarrow$ top; liquid flows top $\rightarrow$ bottom. The column has N computational stages of height $\Delta z = H/N$, where H is the packed height.

2 Chemical Reaction System

2.1 Primary Absorption Reactions

When CO_2 contacts aqueous MEA (RNH_2), two parallel reaction pathways occur depending on the CO_2 loading α (mol CO_2 per mol MEA):

2.1.1 Carbamate Route (dominant at low loading, $\alpha < 0.40$)

$$\mathrm{CO_2 + 2\,RNH_2 \longrightarrow RNHCOO^- + RNH_3^+} \tag{1}$$

This zwitterion-intermediate reaction is overall second-order: first-order in CO_2 and first-order in free MEA. The stoichiometry consumes **two moles of MEA per mole of CO_2**.

2.1.2 Bicarbonate Route (significant at high loading, $\alpha > 0.40$)

$$\mathrm{CO_2 + RNH_2 + H_2O \longrightarrow RNH_3^+ + HCO_3^-} \tag{2}$$

This pathway consumes **one mole of MEA per mole of CO_2**.

2.2 Free MEA Concentration

The concentration of unreacted (free) MEA available for absorption is a function of loading α. Accounting for the weighted stoichiometry:

$$\boxed{C_{\mathrm{MEA,free}} = \max\Big(1 - 2\alpha,\ 0.05\Big) \cdot \frac{w_{\mathrm{MEA}}\,\rho_L}{M_{\mathrm{MEA}}}} \tag{3}$$

where $w_{\mathrm{MEA}} = 0.30$ is the MEA mass fraction, ρ_L is the liquid density, and $M_{\mathrm{MEA}} = 61.08\,\mathrm{g\,mol^{-1}}$. The factor $(1-2\alpha)$ represents the carbamate stoichiometry; the floor at 0.05 prevents numerical singularities near complete loading.

3 Physical and Transport Properties

3.1 Liquid Density

From Weiland et al. (1998), the density of 30 wt% aqueous MEA is fitted as:

$$\boxed{\rho_L(T) = 1060 - 0.57\,(T_{\mathrm{C}} - 25) \quad [\mathrm{kg\,m^{-3}}]} \tag{4}$$

where T_{C} is temperature in °C. This gives $\rho_L = 1051.5\,\mathrm{kg\,m^{-3}}$ at 40 °C, consistent with experimental data.

3.2 Dynamic Viscosity

The Arrhenius-form fit to Weiland et al. (1998) Table 3 data for 30 wt% MEA:

$$\boxed{\mu_L(T,\alpha) = \exp\left(\frac{2121}{T} - 6.40\right) \cdot \big(1 + 0.50\,\alpha\big) \quad [\mathrm{mPa \cdot s}]} \tag{5}$$

The pre-exponential loading correction $(1 + 0.50\alpha)$ accounts for the increased viscosity of loaded solution due to carbamate and bicarbonate ions. At 40 °C and $\alpha = 0.30$: $\mu_L = 1.67\,\mathrm{mPa \cdot s}$.

3.3 Liquid-Phase CO_2 Diffusivity

The Wilke–Chang (1955) equation estimates the diffusivity of CO_2 in the mixed MEA–water solvent. For a binary solvent, the effective parameters are:

$$x_{\mathrm{MEA}} = \frac{w_{\mathrm{MEA}}/M_{\mathrm{MEA}}}{w_{\mathrm{MEA}}/M_{\mathrm{MEA}} + (1 - w_{\mathrm{MEA}})/M_{\mathrm{H_2O}}} \tag{6}$$

$$M_{\mathrm{mix}} = x_{\mathrm{MEA}}\, M_{\mathrm{MEA}} + (1 - x_{\mathrm{MEA}})\, M_{\mathrm{H_2O}} \tag{7}$$

$$\varphi = 2.6\,(1 - x_{\mathrm{MEA}}) + 1.5\, x_{\mathrm{MEA}} \tag{8}$$

where φ is the association parameter ($\varphi_{\mathrm{H_2O}} = 2.6$, $\varphi_{\mathrm{MEA}} = 1.5$). The Wilke–Chang equation then gives:

$$\boxed{D_L = 7.4 \times 10^{-8}\, \frac{\sqrt{\varphi\, M_{\mathrm{mix}}} \cdot T}{\mu_L^{\mathrm{cP}} \cdot V_{\mathrm{CO_2}}^{0.6}} \quad [\mathrm{cm^2\,s^{-1}}]} \tag{9}$$

where $V_{\mathrm{CO_2}} = 34.0\,\mathrm{cm^3\,mol^{-1}}$ is the Le Bas molar volume of CO_2, and μ_L^{cP} is viscosity in cP. Converting: $D_L \times 10^{-4}$ gives $\mathrm{m^2\,s^{-1}}$. At 40 °C: $D_L = 1.26 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$.

3.4 MEA Diffusivity (for Enhancement Factor)

The MEA diffusivity is estimated via the Stokes–Einstein relation relative to CO_2:

$$D_{\mathrm{MEA}} = 1.0 \times 10^{-9} \cdot \frac{313}{T} \cdot \frac{1.5 \times 10^{-3}}{\mu_L} \quad [\mathrm{m^2\,s^{-1}}] \tag{10}$$

3.5 Gas-Phase CO_2 Diffusivity

From the Chapman–Enskog kinetic theory:

$$D_G = 1.6 \times 10^{-5} \left(\frac{T}{273.15}\right)^{1.75} \cdot \frac{P_0}{P} \quad [\mathrm{m^2\,s^{-1}}] \tag{11}$$

where $P_0 = 101.325\,\mathrm{kPa}$ is the reference pressure.

4 Vapour–Liquid Equilibrium (VLE)

4.1 Physical Basis

The chemical equilibrium of CO_2 in MEA solution involves multiple simultaneous ionic equilibria. These are collapsed into a single semi-empirical expression for the equilibrium CO_2 partial pressure p^* as a function of loading α and temperature T.

4.2 Kent–Eisenberg Log-Linear Form

The adopted form is:

$$\boxed{\log_{10} p^*(\alpha, T) = c_0 + c_1\,\alpha + \frac{c_2 + c_3\,\alpha}{T}} \tag{12}$$

where p^* is in kPa and T is in Kelvin. The four coefficients $[c_0, c_1, c_2, c_3]$ are determined by solving the 4×4 linear system formed from the Jou et al. (1995) data:

$$\mathbf{X}\,\mathbf{c} = \mathbf{b} \tag{13}$$

$$\mathbf{X} = \begin{pmatrix} 1 & \alpha_1 & 1/T_1 & \alpha_1/T_1 \\ 1 & \alpha_2 & 1/T_2 & \alpha_2/T_2 \\ 1 & \alpha_3 & 1/T_3 & \alpha_3/T_3 \\ 1 & \alpha_4 & 1/T_4 & \alpha_4/T_4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} \log_{10} p_1^* \\ \log_{10} p_2^* \\ \log_{10} p_3^* \\ \log_{10} p_4^* \end{pmatrix} \tag{14}$$

4.2.1 Calibration Data (Jou et al. 1995)

Point	α	T (°C)	p^* (kPa)	Physical meaning
1	0.25	40	0.30	Absorber lean end
2	0.40	40	3.50	Absorber rich end
3	0.25	120	40.0	Stripper lean end
4	0.45	120	200.0	Stripper rich end

This formulation achieves 0% error at all four calibration points and interpolates smoothly across the absorber operating range ($\alpha = 0.10$–0.55, $T = 40$–$120\,°\mathrm{C}$).

4.3 Physical Henry's Law Constant

The physical (unreacted) solubility of CO_2 in the MEA solution, used in the enhancement factor calculation, follows the Versteeg & van Swaaij (1988) correlation based on the N_2O analogy:

$$\boxed{H_{\mathrm{phys}}(T) = 3.4 \times 10^{-2} \exp\left[2044\left(\frac{1}{298.15} - \frac{1}{T}\right)\right] \quad [\mathrm{kPa \cdot m^3\,mol^{-1}}]} \tag{15}$$

The interface CO_2 concentration is then:

$$C_{CO_2,i} = \frac{p^*(\alpha, T)}{H_{\text{phys}}(T)} \quad [\text{mol}\,\text{m}^{-3}] \tag{16}$$

5 Reaction Kinetics

5.1 Second-Order Rate Constant

For the CO_2–MEA reaction, the Arrhenius expression from Versteeg et al. (1996) specifically fitted to MEA data is:

$$\boxed{k_2(T) = \frac{9.77 \times 10^{10}}{10^3} \exp\left(-\frac{6975}{T}\right) \quad [\text{m}^3\,\text{mol}^{-1}\,\text{s}^{-1}]} \tag{17}$$

The factor 10^3 converts from the original $\text{m}^3\,\text{kmol}^{-1}\,\text{s}^{-1}$ units of Versteeg et al. (1996).

5.2 Pseudo-First-Order Approximation

When the MEA concentration greatly exceeds the CO_2 interface concentration ($C_{\text{MEA,free}} \gg C_{CO_2,i}$), the reaction behaves as pseudo-first-order in CO_2:

$$r_{CO_2} = k_2\, C_{\text{MEA,free}}\, C_{CO_2} = k_{\text{app}}\, C_{CO_2} \tag{18}$$

where the apparent first-order rate constant is:

$$\boxed{k_{\text{app}} = k_2(T)\, C_{\text{MEA,free}} \quad [\text{s}^{-1}]} \tag{19}$$

6 Two-Film Mass Transfer Theory

6.1 Fundamental Premise

The two-film theory (Lewis & Whitman 1924) postulates that all resistance to interphase mass transfer is concentrated in thin stagnant films on either side of the gas–liquid interface. Concentration profiles are linear within each film, and equilibrium holds exactly at the interface.

The molar flux of CO_2 from gas to liquid is:

$$N_{CO_2} = k_G \left(p_{CO_2} - p_i^*\right) = k_L\, E \left(C_{CO_2,i} - C_{CO_2,L}\right) \tag{20}$$

where k_G is the gas-film mass transfer coefficient ($\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}\,\text{kPa}^{-1}$), k_L is the physical liquid-film coefficient ($\text{m}\,\text{s}^{-1}$), and E is the enhancement factor accounting for chemical reaction.

6.2 Overall Gas-Phase Driving Force

Eliminating the interface conditions between the two films gives the overall flux in terms of measurable bulk quantities:

$$N_{CO_2} = K_{OG} \left(p_{CO_2} - p^*\right) \tag{21}$$

where K_{OG} is the overall gas-phase mass transfer coefficient. From resistances-in-series:

$$\frac{1}{K_{OG}} = \frac{1}{k_G} + \frac{1}{k_L^{\text{eff}}} \tag{22}$$

The effective liquid-side coefficient in pressure-driving-force units is:

$$k_L^{\text{eff}} = \frac{E\, k_L^0}{H_{\text{phys}}} \tag{23}$$

where k_L^0 is the physical liquid-film coefficient and H_{phys} is the Henry constant (eq. (15)).

6.3 Driving Force Sign Convention

Absorption occurs when the actual gas-phase partial pressure exceeds the equilibrium value:

$$\Delta p = \max\bigl(p_{\text{CO}_2} - p^*(\alpha, T),\ 0\bigr) \tag{24}$$

A positive Δp drives CO_2 from gas phase into liquid; a negative value would indicate stripping (which is suppressed to zero in the absorber).

7 Mass Transfer Coefficient Correlations

7.1 Gas-Film Coefficient: Bravo & Fair (1982)

For structured and random packings, Bravo & Fair (1982) adapted the Onda et al. (1968) correlation:

$$Re_G = \frac{\rho_G\, u_G\, d_p}{\mu_G} \tag{25}$$

$$Sc_G = \frac{\mu_G}{\rho_G\, D_G} \tag{26}$$

$$Sh_G = 0.407\, Re_G^{0.655}\, Sc_G^{1/3} \tag{27}$$

The gas-film coefficient in $\mathrm{m\,s^{-1}}$ is:

$$k_G^{\text{m/s}} = \frac{Sh_G\, D_G}{d_p} \tag{28}$$

Converting to the pressure-driving-force basis using the ideal gas law ($p = C_G RT$):

$$\boxed{k_G = \frac{k_G^{\text{m/s}}}{RT} \quad [\mathrm{kmol\,m^{-2}\,s^{-1}\,kPa^{-1}}]} \tag{29}$$

where $d_p = 0.025\,\mathrm{m}$ is the equivalent packing diameter.

7.2 Liquid-Film Coefficient: Billet & Schultes (1999)

Billet & Schultes (1999) developed packing-specific correlations from a large experimental database. For Mellapak 250Y structured packing:

$$\boxed{k_L^0 = C_L \sqrt{\frac{D_L\, u_L}{d_h}} \quad [\mathrm{m\,s^{-1}}]} \tag{30}$$

where:

$$C_L = 1.334 \quad \text{(Mellapak 250Y packing constant)}$$
$$d_h = \frac{4\,\varepsilon}{a_p} = \frac{4 \times 0.97}{250} = 0.01552\,\text{m} \quad \text{(hydraulic diameter)}$$
$$u_L = \frac{G_L}{\rho_L} \quad [\text{m}\,\text{s}^{-1}] \quad \text{(superficial liquid velocity)}$$

Here $\varepsilon = 0.97$ is the packing void fraction and $a_p = 250\,\text{m}^2\,\text{m}^{-3}$ is the specific packing area.

The Billet–Schultes correlation physically expresses the liquid-film coefficient as proportional to the square root of liquid diffusivity and velocity — consistent with Higbie (1935) penetration theory: $k_L^0 \propto \sqrt{D_L\,u_L/L_{\text{char}}}$.

8 Chemical Enhancement Factor

8.1 Physical Basis

Without chemical reaction, dissolved CO_2 must diffuse from the interface into the bulk liquid against a concentration gradient. With fast chemical reaction, CO_2 is consumed near the interface, steepening the concentration gradient and thus enhancing the flux beyond what physical absorption alone would give. The enhancement factor E is defined as:

$$E = \frac{\text{flux with reaction}}{\text{flux without reaction (same driving force)}} \geq 1 \tag{31}$$

8.2 Hatta Number

The Hatta number (Ha) is the fundamental dimensionless group characterising the relative rates of chemical reaction and liquid-phase diffusion:

$$\boxed{Ha = \frac{\sqrt{k_{\text{app}}\,D_L}}{k_L^0} = \frac{\sqrt{k_2\,C_{\text{MEA,free}}\,D_L}}{k_L^0}} \tag{32}$$

Physical interpretation:

- $Ha \ll 1$: Slow reaction (film not depleted; $E \approx 1$, physical absorption)
- $Ha \approx 1$–3: Intermediate regime
- $Ha \gg 3$: Fast reaction (pseudo-first-order; $E \approx Ha$)
- $Ha \to \infty$: Instantaneous reaction ($E \to E_\infty$)

For 30 wt% MEA at 40 °C: $Ha \approx 6$–10, confirming the fast pseudo-first-order regime.

8.3 Instantaneous Enhancement Factor

The upper bound on E occurs when reaction is instantaneous (Danckwerts 1970):

$$\boxed{E_\infty = 1 + \frac{D_{\text{MEA}}\,C_{\text{MEA,free}}}{2\,D_L\,C_{\text{CO}_2,i}}} \tag{33}$$

The stoichiometric factor 2 in the denominator reflects the carbamate reaction (eq. (1)): two MEA molecules consumed per CO_2 molecule. $C_{\text{CO}_2,i}$ is obtained from eq. (16).

8.4 DeCoursey (1974) Explicit Approximation

DeCoursey (1974) derived an explicit, closed-form approximation to the Hatta–Danckwerts penetration-theory model that bridges all regimes from slow to instantaneous:

$$\boxed{E = -\frac{Ha^2}{2E_\infty} + \sqrt{\frac{Ha^4}{4E_\infty^2} + Ha^2 + 1}} \tag{34}$$

Limiting cases:

$$E \xrightarrow{Ha \to 0} 1 \quad \text{(no enhancement, physical absorption)}$$

$$E \xrightarrow{Ha \gg 1,\ Ha \ll E_\infty} Ha \quad \text{(fast pseudo-first-order)}$$

$$E \xrightarrow{Ha \gg E_\infty} E_\infty \quad \text{(instantaneous reaction)}$$

The result is bounded: $E = \min(E_{\text{DeCoursey}},\ E_\infty)$.

8.5 Derivation Sketch

The DeCoursey result arises from the penetration theory model for simultaneous diffusion and second-order reaction. Starting from the species continuity equation for CO_2 in the liquid film (1-D, steady-state, fast reaction):

$$D_L \frac{\mathrm{d}^2 C_A}{\mathrm{d}z^2} = k_2\, C_A\, C_B \tag{35}$$

where $C_A = [CO_2]$ and $C_B = [\text{MEA}_{\text{free}}]$. In the pseudo-first-order limit ($C_B \approx C_{\text{MEA,free}} = \text{const}$), this reduces to:

$$D_L \frac{\mathrm{d}^2 C_A}{\mathrm{d}z^2} = k_{\text{app}}\, C_A \tag{36}$$

with boundary conditions $C_A(0) = C_{A,i}$ (interface) and $C_A(\delta_L) = C_{A,L}$ (bulk). The solution is:

$$C_A(z) = C_{A,i} \frac{\sinh\big[Ha\,(1 - z/\delta_L)\big]}{\sinh(Ha)} + C_{A,L} \frac{\sinh\big[Ha\, z/\delta_L\big]}{\sinh(Ha)} \tag{37}$$

The flux at $z = 0$ gives $E = Ha/\tanh(Ha)$ in the pure pseudo-first-order limit. DeCoursey extended this to finite E_∞ via the approximation in eq. (34), valid for all Ha and all E_∞.

9 Absorber Column Model

9.1 Operating Line: Material Balance

For a counter-current absorber operating at steady state, a material balance on CO_2 between the top of the column (stage N) and an arbitrary cross-section gives the **operating line**:

$$F_{\text{inert}} \frac{y}{1-y} + F_{\text{MEA}}\, \alpha = \text{const} \tag{38}$$

where:

$$F_{\text{inert}} = \frac{G_{\text{gas}}\,(1 - y_{\text{in}})}{M_{\text{gas}}} \quad [\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}] \tag{39}$$

$$F_{\text{MEA}} = \frac{G_L\, w_{\text{MEA}}}{M_{\text{MEA}}} \quad [\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}] \tag{40}$$

Here G_{gas} and G_L are the superficial mass fluxes ($\text{kg}\,\text{m}^{-2}\,\text{s}^{-1}$), and $M_{\text{gas}} = 0.030\,\text{kg}\,\text{mol}^{-1}$ is the approximate mean molecular weight of the inert gas.

Applying the operating line between an arbitrary stage (gas mole fraction y) and the top (gas mole fraction y_{top}, loading α_{lean}):

$$\boxed{\alpha(y) = \alpha_{\text{lean}} + \frac{F_{\text{inert}}}{F_{\text{MEA}}}\left[\frac{y}{1-y} - \frac{y_{\text{top}}}{1-y_{\text{top}}}\right]} \tag{41}$$

This equation *analytically* couples the liquid loading to the gas composition at every cross-section, eliminating one degree of freedom and making the column integration well-posed.

9.2 Stage Mass Balance: Gas Phase

For stage j (height $\Delta z = H/N$), the molar balance on CO_2 in the gas phase is:

$$\underbrace{F_{\text{inert}}\,\frac{y_j}{1-y_j}}_{\text{CO}_2\text{ in}} - \underbrace{N_{\text{CO}_2,j}\,a_p\,\Delta z}_{\text{absorbed}} = \underbrace{F_{\text{inert}}\,\frac{y_{j+1}}{1-y_{j+1}}}_{\text{CO}_2\text{ out}} \tag{42}$$

Rearranging for the outlet mole fraction:

$$\boxed{y_{j+1} = \frac{n_{\text{out}}}{F_{\text{inert}} + n_{\text{out}}}} \quad \text{where} \quad n_{\text{out}} = \max\left(\frac{y_j}{1-y_j}\,F_{\text{inert}} - N_{\text{CO}_2,j}\,a_p\,\Delta z,\ 0\right) \tag{43}$$

and $N_{\text{CO}_2,j} = K_{OG}\,\Delta p_j$ from eq. (21).

9.3 Stage Energy Balance: Liquid Temperature

Heat released by exothermic absorption raises the liquid temperature. The energy balance on the liquid across stage j (flowing top to bottom):

$$G_L\,C_{p,L}\,\frac{\mathrm{d}T_L}{\mathrm{d}z} = -N_{\text{CO}_2}\,a_p\,(-\Delta H_{\text{abs}}) \tag{44}$$

The negative sign on ΔH_{abs} converts from the exothermic convention. Discretised over stage height Δz:

$$\boxed{T_{L,j} = T_{L,j+1} + \frac{\Delta H_{\text{abs}}\,N_{\text{CO}_2,j}\,a_p\,\Delta z}{G_L\,C_{p,L}}} \tag{45}$$

where $\Delta H_{\text{abs}} = 85\,\text{kJ}\,\text{mol}^{-1}$ (exothermic heat of CO_2 absorption into MEA, Kim & Svendsen 2007) and $C_{p,L} = 3800\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$ is the heat capacity of the solvent.

9.4 Capture Rate

The fractional CO_2 capture efficiency is defined as:

$$\boxed{\eta = 1 - \frac{y_{\text{top}}}{y_{\text{in}}} \quad [\text{dimensionless, expressed as }\%]} \tag{46}$$

9.5 Rich Loading

From the operating line (eq. (41)) evaluated at the column bottom ($y = y_{\text{in}}$):

$$\alpha_{\text{rich}} = \alpha_{\text{lean}} + \frac{F_{\text{inert}}}{F_{\text{MEA}}} \left[\frac{y_{\text{in}}}{1 - y_{\text{in}}} - \frac{y_{\text{top}}}{1 - y_{\text{top}}} \right] \tag{47}$$

10 Stripper and Reboiler Energy Balance

10.1 Overview

The stripper (regenerator) reverses the absorption reactions by heating the rich solvent. The reboiler provides the thermal energy required. The lean loading exiting the stripper depends on how much CO_2 can be desorbed with the available reboiler duty Q_{reb}.

10.2 Reboiler Energy Balance

The total reboiler duty is partitioned into three contributions (Fregúia & Rochelle 2003; Abu-Zahra et al. 2007):

$$\boxed{Q_{\text{reb}} = Q_{\text{des}} + Q_{\text{sens}} + Q_{\text{steam}}} \tag{48}$$

10.2.1 Desorption Heat

Energy required to break the chemical bonds formed during absorption:

$$Q_{\text{des}} = \dot{n}_{CO_2} \, \Delta H_{\text{des}} \tag{49}$$

where $\Delta H_{\text{des}} = 85\,\text{kJ}\,\text{mol}^{-1}$ (endothermic) and $\dot{n}_{CO_2}$ is the molar CO_2 stripping rate:

$$\dot{n}_{CO_2} = F_{\text{MEA}} \left(\alpha_{\text{rich}} - \alpha_{\text{lean}} \right) \quad [\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}] \tag{50}$$

10.2.2 Sensible Heat (After Rich–Lean Heat Exchanger)

In the complete process, a rich–lean heat exchanger (HEX) pre-heats the rich solvent before entry to the stripper, and pre-cools the lean solvent before recycle to the absorber. After the HEX, the solvent must be raised from approximately 100 °C to the reboiler temperature (120 °C), a lift of $\Delta T_{\text{sens}} \approx 20\,\text{K}$:

$$Q_{\text{sens}} = G_L \, C_{p,L} \, \Delta T_{\text{sens}} \tag{51}$$

10.2.3 Steam Overhead (Water Vaporisation)

The stripping steam required to maintain the vapour partial pressure of CO_2 in the stripper overhead:

$$Q_{\text{steam}} = r_{H_2O} \, \dot{m}_{CO_2} \, \lambda_{\text{steam}} \tag{52}$$

where $r_{H_2O} = 0.45\,\text{kg}_{H_2O}/\text{kg}_{CO_2}$ is the ratio of water evaporated to CO_2 stripped, and $\lambda_{\text{steam}} = 2200\,\text{kJ}\,\text{kg}^{-1}$ is the latent heat of vaporisation of water.

10.3 Specific Energy Consumption

The primary performance metric for the regeneration section is the **specific reboiler duty**, normalised by the mass of CO_2 captured:

$$\boxed{\epsilon = \frac{Q_{\text{reb}}}{\dot{m}_{\text{CO}_2}} \quad \left[\text{GJ tonne}_{\text{CO}_2}^{-1}\right]} \tag{53}$$

where $\dot{m}_{\text{CO}_2} = \dot{n}_{\text{CO}_2} M_{\text{CO}_2}$ ($\text{kg m}^{-2}\,\text{s}^{-1}$) and Q_{reb} is in W m^{-2}. Typical literature values for 30 wt% MEA at 90% capture: $\epsilon = 3.2$–$4.5\,\text{GJ tonne}^{-1}$ (Abu-Zahra et al. 2007).

11 Packing Parameters and Column Geometry

The simulation uses Mellapak 250Y structured metal packing with the following geometric parameters:

Parameter	Symbol	Value
Specific packing area	a_p	$250\,\text{m}^2\,\text{m}^{-3}$
Void fraction	ε	0.97
Hydraulic diameter	$d_h = 4\varepsilon/a_p$	0.01552 m
Packing height	H	15 m
Number of stages	N	60
Stage height	$\Delta z = H/N$	0.25 m
Billet–Schultes constant	C_L	1.334
Equivalent particle diameter	d_p	0.025 m

12 Summary of Model Equations

12.1 Equation Hierarchy

The complete model solves the following set of equations at each stage j of the absorber:

1. **Operating line** (eq. (41)): $\alpha_j = f(y_j, y_{\text{top}})$
2. **Physical properties**: ρ_L, μ_L, D_L, D_{MEA} from eqs. (4) to (10)
3. **Free MEA**: $C_{\text{MEA,free}}$ from eq. (3)
4. **VLE**: $p^*(\alpha_j, T_{L,j})$ from eq. (12)
5. **Liquid MTC**: k_L^0 from eq. (30)
6. **Hatta number**: Ha_j from eq. (32)
7. **Enhancement factor**: E_j from eq. (34)
8. **Gas MTC**: $k_{G,j}$ from eq. (29)
9. **Overall MTC**: $K_{OG,j}$ from eqs. (22) and (23)
10. **Driving force**: Δp_j from eq. (24)
11. **CO_2 flux**: $N_{\text{CO}_2,j} = K_{OG,j}\,\Delta p_j$
12. **Gas update**: y_{j+1} from eq. (43)

13. **Temperature update**: $T_{L,j}$ from eq. (45)

The stripper then evaluates eqs. (48) and (53) using α_{rich} from eq. (47).

12.2 Simplifications and Assumptions

A1. Steady state: All time derivatives are zero. Applicable to simulation of equilibrium operating points for dataset generation.

A2. Plug flow: Both gas and liquid streams flow as plugs with no axial mixing. Validated for structured packings with $L/D > 5$.

A3. Isothermal gas: Gas-phase temperature variations are small compared to liquid-phase variations. Gas is assumed isothermal at the local liquid temperature.

A4. Henry's law at interface: Chemical equilibrium holds exactly at the gas–liquid interface.

A5. Pseudo-first-order reaction: The free MEA concentration is taken as approximately uniform within the liquid film ($C_{\text{MEA,free}} \gg C_{\text{CO}_2,i}$), which is satisfied for $\alpha < 0.50$.

A6. Constant heat of absorption: $\Delta H_{\text{abs}} = 85\,\text{kJ}\,\text{mol}^{-1}$ is taken as temperature- and loading-independent over the absorber operating range..

13 Model Validation and Parameter Values

13.1 VLE Validation

α	T (°C)	p^*_{Jou} (kPa)	p^*_{model} (kPa)	Error (%)
0.25	40	0.30	0.30	< 1
0.40	40	3.50	3.50	< 1
0.25	120	40.0	40.0	< 1
0.45	120	200.0	200.0	< 1

13.2 Property Validation at Nominal Conditions

($\alpha = 0.30$, $T = 40\,°\text{C}$, 30 wt% MEA)

Property	Model	Literature	Source
ρ_L (kg m^{-3})	1051.5	1040–1060	Weiland (1998)
μ_L (mPa·s)	1.67	1.5–2.0	Weiland (1998)
D_L (m^2 s^{-1})	1.26×10^{-9}	$\sim 1 \times 10^{-9}$	Versteeg (1988)
Ha	8.9	5–15	Cents (2005)
E	8.2	$\sim Ha$ (fast regime)	Aboudheir (2003)
Capture rate (%)	97.7	85–99	Pilot plants
α_{rich}	0.458	0.40–0.50	Pilot plants
$T_{L,\text{bot}}$ (°C)	54.2	50–70	Pilot plants
ϵ (GJ/tonne)	4.62	3.5–5.0	Abu-Zahra (2007)

13.3 Output Variables

Output	Symbol
CO_2 capture rate	η (%)
Rich loading	α_{rich}
Liquid temperature at bottom	$T_{L,\text{bot}}$ (°C)
Exit CO_2 mole fraction	$y_{CO_2,\text{out}}$
Specific reboiler duty	ϵ (GJ/tonne)
Average Hatta number	$\overline{Ha}$
Average enhancement factor	$\bar{E}$

14 References

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